

# Design of a 400 Gb/s Silicon Photonics Transceiver

## For IEEE 802.3 DR4 Intra-DC Interconnects

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# Motivation and Targets

- Intra-datacenter PAM-4 receivers need **high responsivity**, **very low dark current**, and **bandwidth above 100 GHz**.
- A **Ge-on-Si vertical n-i-p photodiode** offers a strong fit with standard SOI photonics.
- The chapter benchmarks the design against Yang Shi et al., *Photonics Research* 12(1), 2024.

| Metric       | Target                                   |
|--------------|------------------------------------------|
| Responsivity | $\geq 0.95 \text{ A W}^{-1}$             |
| Dark current | $\leq 1.3 \text{ nA}$                    |
| Bandwidth    | $\geq 103 \text{ GHz}$                   |
| Detectivity  | $\geq 2.95 \times 10^{10} \text{ Jones}$ |

- Vertical **n-i-p** stack on a 220 nm **SOI** platform.
- 50 nm **n++ Ge** top contact layer.
- 350 nm **intrinsic Ge** absorber.
- **p++ Si** platform for the anode path.
- **U-shaped electrode** reduces the RC penalty while preserving carrier collection.

| Layer  | Thickness |
|--------|-----------|
| n++ Ge | 50 nm     |
| i-Ge   | 350 nm    |
| p++ Si | 220 nm    |
| BOX    | 2 $\mu$ m |

# Optical Simulation Flow

- **Lumerical FDTD** models the taper-coupled optical absorption at 1310 nm.
- The simulation exports **optical generation rate** into CHARGE.
- Ge length is optimized to reach high absorption with minimum footprint.

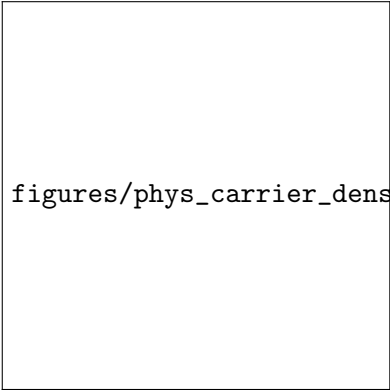
## Key optical design points:

- TE-driven O-band operation
- Adiabatic spot-size converter (500 nm  $\rightarrow$  5  $\mu$ m over 40  $\mu$ m)
- 3D generation-rate export:  $G = P_{\text{abs}}/(h\nu)$

figures/opt\_generation\_rate\_ma

# Electrical Simulation Flow

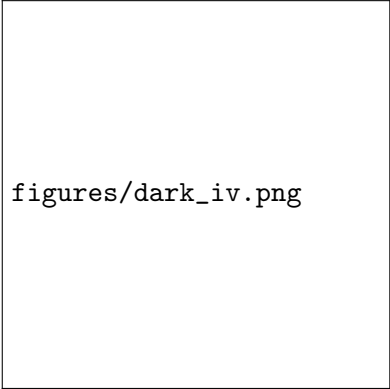
- **Lumerical DEVICE CHARGE** solves carrier transport under reverse bias.
- SRH recombination enabled with  $\tau_n = \tau_p$  (calibrated to match dark current target).
- Fine mesh override: 20 nm max edge length over the Ge active region.
- Extracted quantities include:
  - ▶ Dark current and illuminated current
  - ▶ Band diagram, carrier density
  - ▶ Electric field and potential
- Bandwidth estimated from **transit-time + RC** limits.



figures/phys\_carrier\_density.p

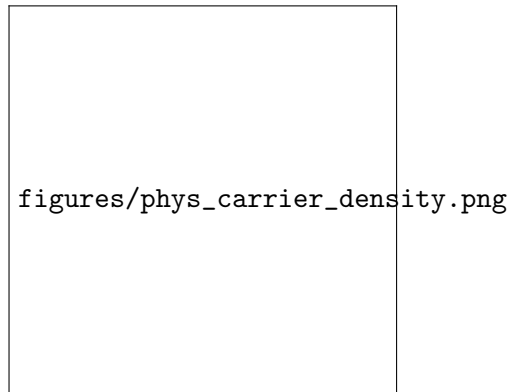
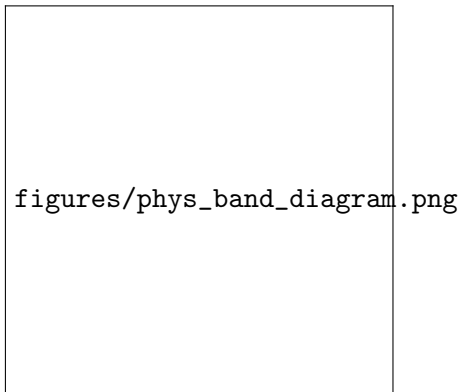
# Dark Current Result

- All parameters flow from solvers—no hardcoded paper values.
- Low dark current is essential for:
  - ▶ Receiver sensitivity
  - ▶ Noise performance
  - ▶ Robust PAM-4 eye opening



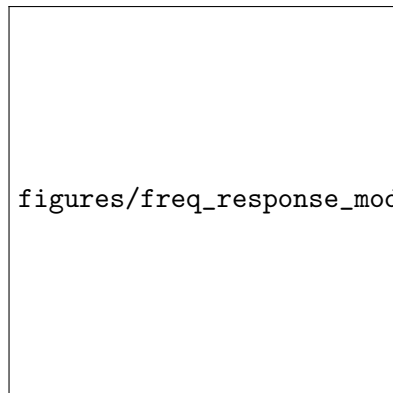
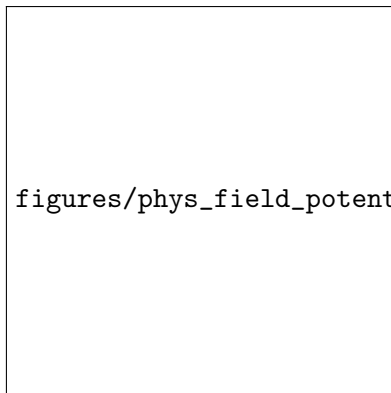
figures/dark\_iv.png

# Internal Device Physics



Band alignment, carrier density, and depletion shaping near  $V = -1$  V explain the tradeoff between responsivity, dark current, and speed.

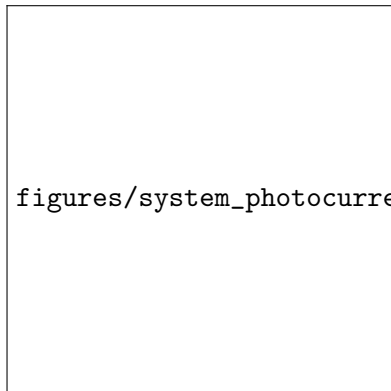
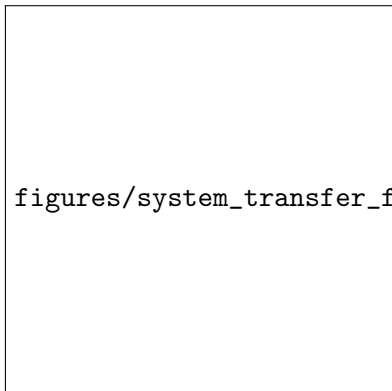
## Electric Field and Bandwidth



- Transit-time and RC contributions are combined into the bandwidth model.
- All parasitic parameters ( $R_s$ ,  $C_j$ ,  $L_p$ ,  $C_p$ ) derived from geometry.



## System-Level Receiver View



- The repo extends the device into a **MATLAB PAM-4 receiver model** at 53.125 GBd.
- All device parameters flow from the CML `.mat` generated by device-level postprocessing.

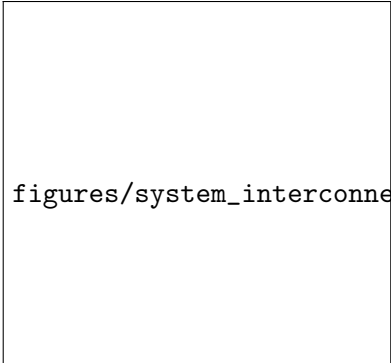
## PAM-4 Signal Quality

figures/system\_pam4\_histogram.png

- Matched-filter output shows **4 distinct PAM-4 levels**.
- Decision thresholds separate the symbol clusters.
- SNR, BER, and SER are computed from the simulation.
- The PD can now be studied both as a device and as a circuit block.

# INTERCONNECT-Ready Compact Model

- The repo exports a **Lumerical INTERCONNECT-ready** photodetector data package.
- Generated artifacts include:
  - ▶ Responsivity table (frequency-dependent)
  - ▶ Bandwidth table (bias-dependent)
  - ▶ Dark-current table
  - ▶ Compact-model .mat source data
  - ▶ Equivalent-circuit parameters CSV



figures/system\_interconnect\_compact\_model.png

# Data Pipeline Overview

FDTD .lsf → fdtdd\_results.mat → CHARGE .lsf → charge\_results.mat → postprocess.m → CML .mat  
→ system-level

All parameters flow from Lumerical solvers → MAT files → MATLAB postprocess → CML export → system-level scripts. **Nothing is hardcoded in the pipeline.**

# Final Takeaways

- The chapter demonstrates a **Ge-on-Si O-band photodiode** targeting published high-speed metrics.
- The workflow is consistent across:
  - ▶ **Device-level FDTD**—optical absorption, generation rate
  - ▶ **Device-level CHARGE**—dark current, responsivity, band structure
  - ▶ **Device-level postprocess**—CML dataset, all derived parameters
  - ▶ **System-level MATLAB**—PAM-4 BER, eye diagrams, histograms
  - ▶ **INTERCONNECT-ready export**—circuit-level integration
- **Zero hardcoded paper values**—all metrics from solvers and geometry.